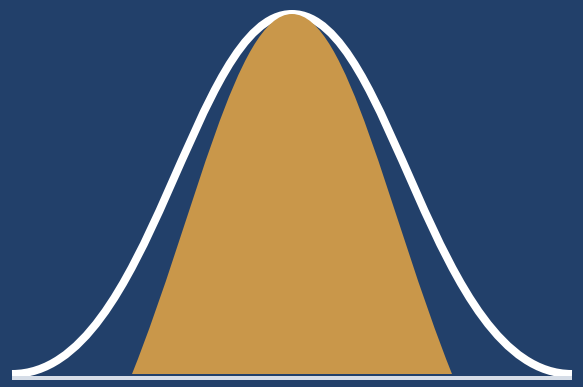
EXCERPT | PARENT ROUNDTABLE SERIES

The Why Behind Testing

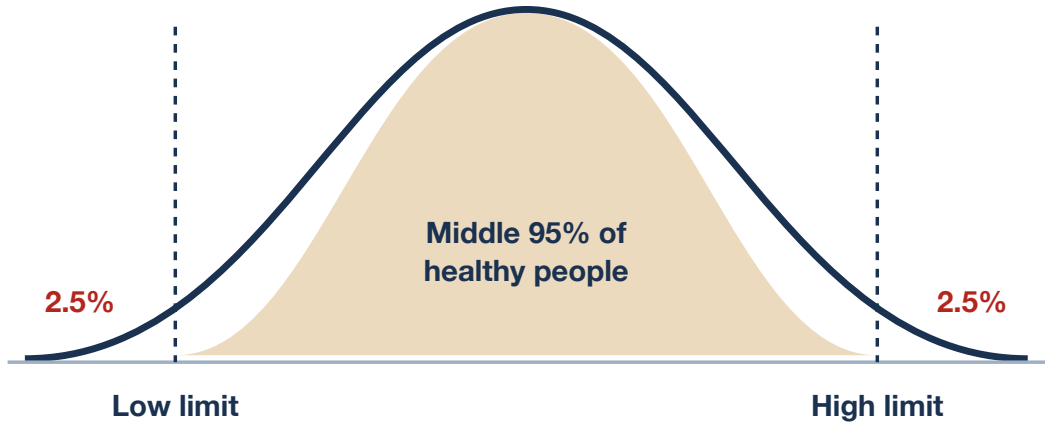
What a lab test measures, how results are read, and what a test cannot tell you

Deborah M. Burke-Allen, RPh
Registered Pharmacist | Parent education roundtable



READING THE RANGE

A reference range describes most healthy people, not ideal health



- Labs set the range from the middle 95% of a healthy reference group
- So about 1 in 20 healthy people land outside the range on any single test
- Ranges vary by lab, age, and method, so compare results from the same lab when you can

WORKED EXAMPLE

Reading a vitamin D result: the number, the range, and the next question

Sample result	Value
25-hydroxy vitamin D	**24 ng/mL**
This lab's range	30 - 100 ng/mL
Endocrine Society deficiency	Below 20 ng/mL
Endocrine Society insufficiency	21 - 29 ng/mL

Sample values for teaching. Not a real patient.

How I would read it

- Insufficient, not deficient
- Next questions: season, sun, diet, absorption, and current supplements
- Make one change, then recheck when the clinician advises

Holick MF, et al. J Clin Endocrinol Metab. 2011;96(7):1911-30. PMID 21646368

PHARMACIST LENS

What a test can miss, and what can skew it

- A normal result does not rule out every problem
- An abnormal result is not a diagnosis on its own
- Illness, hydration, fasting, and time of day can shift values

Medicines and supplements matter

High-dose biotin, common in hair, skin, and nail products, can falsely raise or lower some lab results, including thyroid tests and troponin. Tell the lab about every supplement before testing.

WANT THE FULL DECK?

This is an excerpt from a full parent-education roundtable

- Designed, researched, and presented by a registered pharmacist
- Every clinical claim cited and verified in PubMed
- Written for parents; reviewed to a professional medical-content standard

See the full deck and more

deborahburkeallen.github.io/recruiters.html

bridgestowellness@icloud.com